

PHY F376: A study of hybrid classical-quantum algorithms

Week 5 (18/9/2026) - Troubleshooting Hardware Access via the Classroom Account

Khushi Pandey (2023B5A70951G)

Overview of This Week's Work

This week's goal was to move the SQD workflow from the noiseless simulator onto real IBM hardware, using the instance through Prof. Indrakshi's classroom account. In practice, most of the week went into setting up and troubleshooting API access rather than running anything new, since I was not able to get a working set of credentials, and this has not yet been resolved even after going through it with her.

Setting Up Credentials Securely

Since my code is version-controlled on GitHub, the first part of this was making sure I did not commit any API key or instance identifier in plain text. I set this up so that the token and instance CRN are entered through a hidden prompt at runtime instead of being written into the notebook itself, so nothing sensitive ends up in the repository regardless of what gets pushed.

The Access Issue

Once the credential handling was in place, the actual problem was finding and using a valid instance CRN for the account I was added to. I went through the IBM Quantum Platform dashboard to locate my API key and the instance associated with my Prof. Indrakshi's classroom account, but authentication did not succeed with the credentials I found. I went over the account setup with her directly, and she also expected this to work given how the classroom account was configured, but neither of us was able to identify what was misconfigured or missing.

What Is Already in Place

Separately from the access issue, the workflow itself is not blocked in any code sense: the same script that runs the noiseless simulator already has the real-hardware submission path written, differing only in which Sampler object is passed in, so once valid credentials are available, running on hardware should not require any further changes to the workflow logic itself.

Concluding Remarks

This week did not produce new SQD results, since the hardware run itself is still blocked on an account access issue that is outside of my control and is being looked into. I wanted to record this honestly rather than leave it unaddressed, since resolving this is a prerequisite for the actual hardware comparison this project needs. Once access is sorted out, I expect to be able to run the hardware experiment immediately, since the code side of it is already prepared and tested.